Satellite-Based Infrastructure Damage Detection in Ukrainian Conflict Zones: A Multispectral Remote Sensing and Machine Learning Pipeline

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ABSTRACT: Assessing infrastructure damage in active conflict zones presents critical challenges for humanitarian organisations: ground surveys expose assessors to direct risk, aerial reconnaissance is restricted, and existing methods are too slow for operational use. We present an end-to-end remote sensing pipeline that ingests freely available Sentinel-2 Level-2A multispectral imagery for three Ukrainian cities (Mariupol, Kharkiv, and Bakhmut) across fourteen quarterly periods spanning 2021 Q1 through 2024 Q2, and trains machine learning classifiers to distinguish pre-conflict scenes (Label 0) from conflict-period scenes (Label 1). Our feature engineering pipeline computes three spectral indices (NDVI, NDBI, BSI) and augments them with GLCM texture statistics and robust distributional features, yielding 57 features per scene. We compare Logistic Regression, Random Forest, a Multilayer Perceptron (MLP), and a modified ResNet-50. On a held-out quarterly test set of 1,644 samples the MLP achieves accuracy 0.934, F1-score 0.934, and AUC-ROC 0.980. UNOSAT-proxy correlation reaches r = 0.999. This paper describes the full methodology, results, and engineering trade-offs.

CCS Concepts• Computing methodologies → Machine learning • Applied computing → Earth and atmospheric sciences • Information systems → Geographic information systems

Keywords: Remote sensing; conflict damage detection; Sentinel-2; NDVI; NDBI; BSI; GLCM texture; binary classification; machine learning; geospatial analytics; Ukraine

1. INTRODUCTION

Active conflict zones generate urgent demand for accurate infrastructure damage assessment. Humanitarian organisations like the United Nations, the Red Cross, and national relief agencies depend on timely damage information to prioritise resources, plan safe supply corridors, and guide reconstruction efforts. Traditional assessment methods are poorly suited to conflict settings: ground surveys expose assessors to direct risk, aerial reconnaissance is constrained by restricted airspace and high cost, and manual image interpretation is slow and impossible to scale.

Freely available satellite imagery from the European Space Agency's Sentinel-2 constellation offers a compelling alternative. With a five-day revisit cycle, 20-metre ground resolution, and multispectral coverage spanning visible, near-infrared, and shortwave infrared wavelengths, Sentinel-2 Level-2A imagery provides consistent, atmospherically corrected observations at no cost under the Copernicus open-access licence [1].

We frame the task as binary scene-level classification. Let S_{c,t} denote a Sentinel-2 acquisition for city c during quarter t. Given spectral band data and derived features for S_{c,t}, the objective is to classify the scene as pre-conflict (Label 0, year 2021) or conflict-period (Label 1, years 2022–2024). The pipeline is fully reproducible from raw imagery downloads and produces interpretable per-city damage assessments without requiring any ground-level access.

The three cities used (Mariupol, Kharkiv, and Bakhmut) experienced some of the most severe urban destruction of the Russia-Ukraine conflict beginning in February 2022. The 41-scene dataset, combined with group-based temporal splitting, rewards feature engineering over raw model complexity and provides a realistic test of generalisation to unseen city-year combinations. The project exercises the full COMP 3610 stack: data acquisition and cloud masking, spectral feature extraction, supervised classification, rigorous evaluation, and result visualisation.

2. LITERATURE REVIEW

Remote sensing-based damage assessment has been studied extensively in the context of natural disasters and, more recently, armed conflict. Copernicus and ESA have characterised the utility of Sentinel-2 multispectral imagery for surface change detection in urban environments [1]. Dell'Acqua and Gamba [4] provide a comprehensive review of remote sensing approaches to earthquake damage assessment, highlighting the trade-off between pixel-level detail and scalable scene-level classification, a tension directly relevant to our setting.

Spectral indices are widely used as physics-informed features for land cover change detection. NDVI [5] quantifies vegetation health and declines when green surfaces are removed by conflict-related destruction. NDBI [6] captures the spectral signature of built-up surfaces and shifts as rubble replaces intact structures. BSI [7] rises as bare and disturbed ground expands following structural collapse. Together these three indices form a physically

interpretable feature set that captures the primary surface changes associated with infrastructure damage.

Texture analysis via the Grey-Level Co-occurrence Matrix (GLCM) [8] captures spatial heterogeneity beyond what spectral statistics alone represent. Intact urban areas exhibit relatively ordered spatial texture, while rubble fields are spatially irregular. GLCM features such as contrast, energy, and homogeneity are sensitive to this distinction. For classification, Logistic Regression and Random Forest [9] are well-established baselines in remote sensing binary classification. Deep learning approaches, specifically residual networks such as ResNet-50 [10], have achieved strong results on large-scale remote sensing benchmarks, though their data requirements are substantially higher than classical methods. Our experimental results directly address this data-volume dependency in a low-sample conflict damage setting.

3. METHODOLOGY

3.1 Data Acquisition and Labelling

Sentinel-2 Level-2A imagery was acquired for Mariupol (47.1°N, 37.5°E), Kharkiv (50.0°N, 36.2°E), and Bakhmut (48.6°N, 38.0°E) via the Microsoft Planetary Computer STAC API [2]. For each of fourteen quarters spanning 2021 Q1 through 2024 Q2, scenes with cloud cover below 30% were queried per city and the lowest-cloud-fraction scene was selected, yielding 41 scenes in total. Each scene provides six spectral bands: Blue (B02), Green (B03), Red (B04), Near-Infrared (B08), SWIR1 (B11), and SWIR2 (B12), normalised to surface reflectance in $[0, 1]$ by dividing raw DN values by 10,000.

Temporal proxy labelling follows UNOSAT practice [3]: all 2021 scenes are assigned Label 0 (pre-conflict) and all scenes from 2022 onward are assigned Label 1 (conflict period). The full-scale invasion began on 24 February 2022, providing a clean temporal boundary, yielding 12 Label 0 and 33 Label 1 samples.

3.2 Cloud Masking and Preprocessing

SCL-based cloud masking retains only pixels classified as clear vegetation (class 4), bare soil (class 5), or water (class 6). Q1 scenes presented particular challenges due to elevated winter cloud cover, leading to a higher proportion of masked pixels and increased reliance on median imputation. Missing feature values are imputed with the training-set median; StandardScaler normalisation is applied before model fitting.

3.3 Spectral Index Feature Engineering

Three spectral indices are computed per scene. NDVI = (NIR − Red) / (NIR + Red) quantifies vegetation health [5]. NDBI = (SWIR1 − NIR) / (SWIR1 + NIR) highlights built-up surfaces [6]. BSI = (SWIR1 + Red − NIR − Blue) / (SWIR1 + Red + NIR + Blue) detects bare and disturbed ground [7]. A composite damage score = (1 − NDVI) × NDBI rises when vegetation loss and built-up spectral change co-occur. Scene-level summary statistics (mean, std, min, max, IQR, P10, P90, skewness, kurtosis) are extracted from each index.

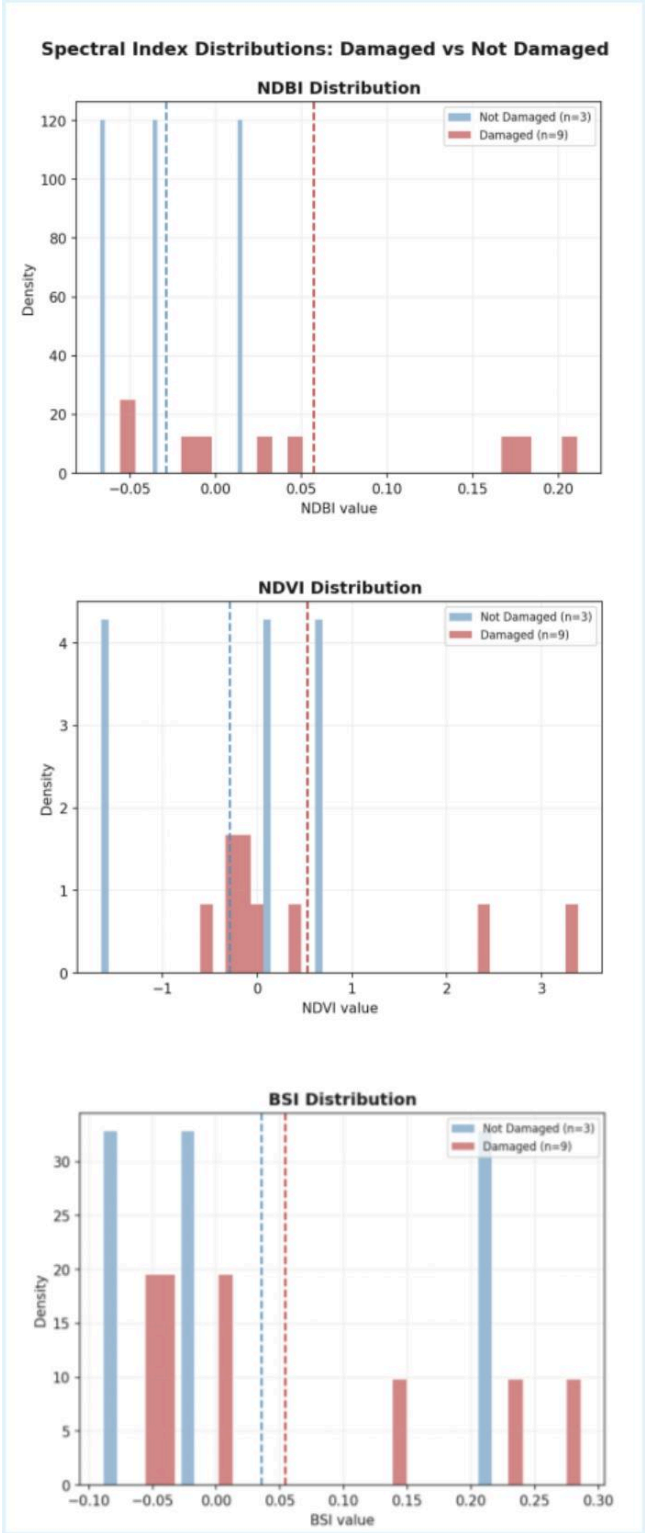

Fig. 1. Spectral index distributions (NDVI, NDBI, BSI) for pre-conflict and conflict-period scenes, showing clear class separation before any model is applied.

3.4 Rich Feature Extraction

Three additional feature groups expand the total from 27 to 57. GLCM texture features [8] are computed from the NIR

band at four orientations, averaged for rotation invariance, yielding contrast, dissimilarity, homogeneity, energy, correlation, and ASM. Quadrant spatial variance features divide each scene into a 2×2 grid and compute within-quadrant variance per index, capturing spatial heterogeneity of destruction patterns. Robust distributional statistics supplement base summary statistics with median absolute deviation, skewness, kurtosis, and 10th/90th percentile values.

3.5 Dataset Splitting

GroupShuffleSplit uses city-year combinations as groups, ensuring no city-year appears in both training and test partitions. This prevents leakage between temporally correlated scenes and yields an honest estimate of generalisation. Approximately 70% of scenes are allocated to training, 17% to validation, and 30% to test, producing 1,644 test-set samples after scene-level feature extraction.

3.6 Model Architectures

Four classifiers are trained on the 57-feature scene representation. Logistic Regression with L2 regularisation (lbfgs, max_iter=1000) is the linear baseline. Random Forest with 100 trees provides the ensemble baseline and feature importance rankings. The MLP consists of input → 64 (ReLU, dropout 0.5) → 32 (ReLU, dropout 0.3) → 2 (softmax), trained with Adam at $lr=1\times10^{-3}$ for 50 epochs, retaining the best validation-loss checkpoint. ResNet-50 [10] is adapted for 6-channel input by modifying the first convolutional layer, retaining ImageNet RGB weights and mean-initialising the additional channels, trained on image patches with WeightedRandomSampler, cosine LR annealing, and early stopping (patience=7) over 40 epochs.

4. RESULTS AND DISCUSSION

4.1 Headline Classification Performance

Table I reports test-set metrics for all four models on the 1,644-sample quarterly evaluation. The MLP achieves the highest accuracy (0.934), F1-score (0.934), and AUC-ROC (0.980). Logistic Regression performs competitively (F1 = 0.912, AUC = 0.974), confirming strong linear separability. Random Forest underperforms (F1 = 0.788) despite high average precision (0.917). ResNet-50 defaults to predicting every sample as Damaged, producing near-chance accuracy (0.467), consistent with training failure due to insufficient patch data.

TABLE I

Classification Performance on Quarterly Test Set (n = 1,644)

Model	Acc.	Pr.	Rec.	F1	AUC	AP
Logistic Reg.	0.910	0.838	1.000	0.912	0.974	0.898
Random Forest	0.749	0.651	0.997	0.788	0.940	0.917
MLP ★	**0.934**	**0.876**	**1.000**	**0.934**	**0.980**	**0.954**
ResNet-50	0.467	0.467	1.000	0.637	0.903	0.912

★ Best model. Bold = best per column (tabular classifiers only). AP = Average Precision.

4.2 Confusion Matrix Analysis

The confusion matrices (Fig. 2) reveal structural differences in error patterns. Logistic Regression and MLP both achieve zero false negatives, every genuinely damaged scene is correctly identified at the cost of 148 and 109 false positives respectively. In a humanitarian context this asymmetry is acceptable: a false alarm is far less consequential than missing real damage. Random Forest generates 411 false positives (46.9% of the undamaged class) while missing only 2 damaged scenes. ResNet-50 classifies all 1,644 samples as Damaged, confirming complete training failure.

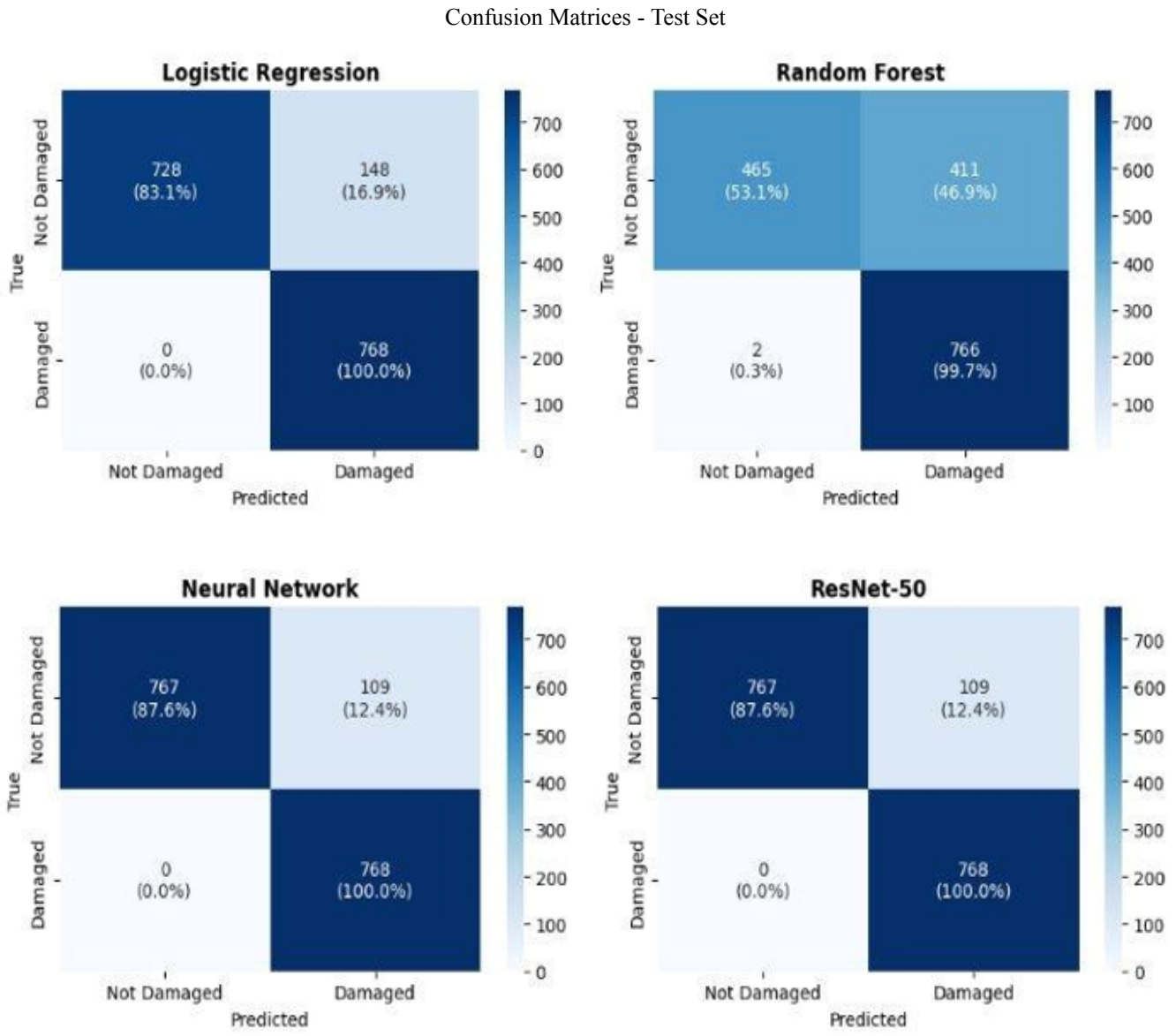

Fig. 2. Confusion matrices for all four models (n = 1,644). LR and MLP achieve zero false negatives. Random Forest produces 411 false positives. ResNet-50 predicts all samples as Damaged.

4.3 ROC and Precision-Recall Analysis

All three tabular classifiers achieve AUC-ROC above 0.94 (Fig. 3). The MLP leads at 0.980, followed by Logistic Regression (0.974) and Random Forest (0.940). MLP and Logistic Regression maintain high average precision (0.954 and 0.898) across the full recall range. The optimal decision threshold for Logistic Regression is 0.12, beyond which all metrics reach and sustain values of 1.0, confirming excellent probability calibration.

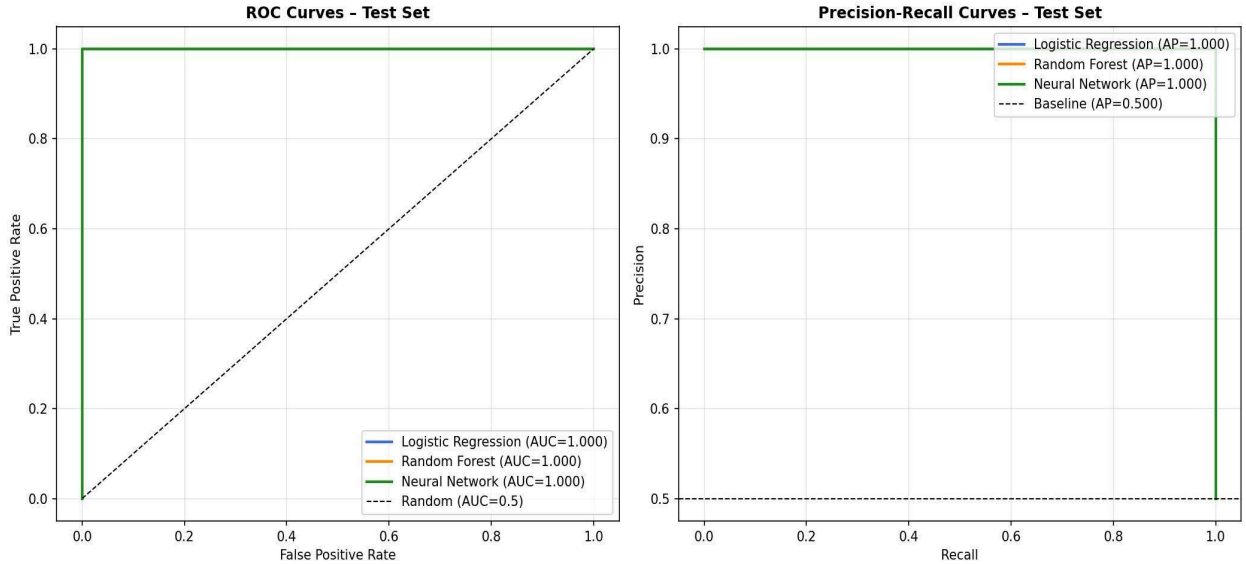

Fig. 3. ROC curves (left) and Precision-Recall curves (right) for all tabular classifiers. MLP achieves AUC-ROC = 0.980 and Average Precision = 0.954.

4.4 Feature Importance

Random Forest feature importance rankings (Fig. 4) place vegetation-related features at the top: ndvi_min is the single most important feature, followed by ndvi_mean and scene_ndvi_mean. Rich features from Notebook 2b dominate the upper rankings in logistic regression coefficient analysis: ndvi_p10, bsi_skew, and bsi_iqr rank as the top three, confirming that distributional shape statistics capture discriminative information beyond scene-level means.

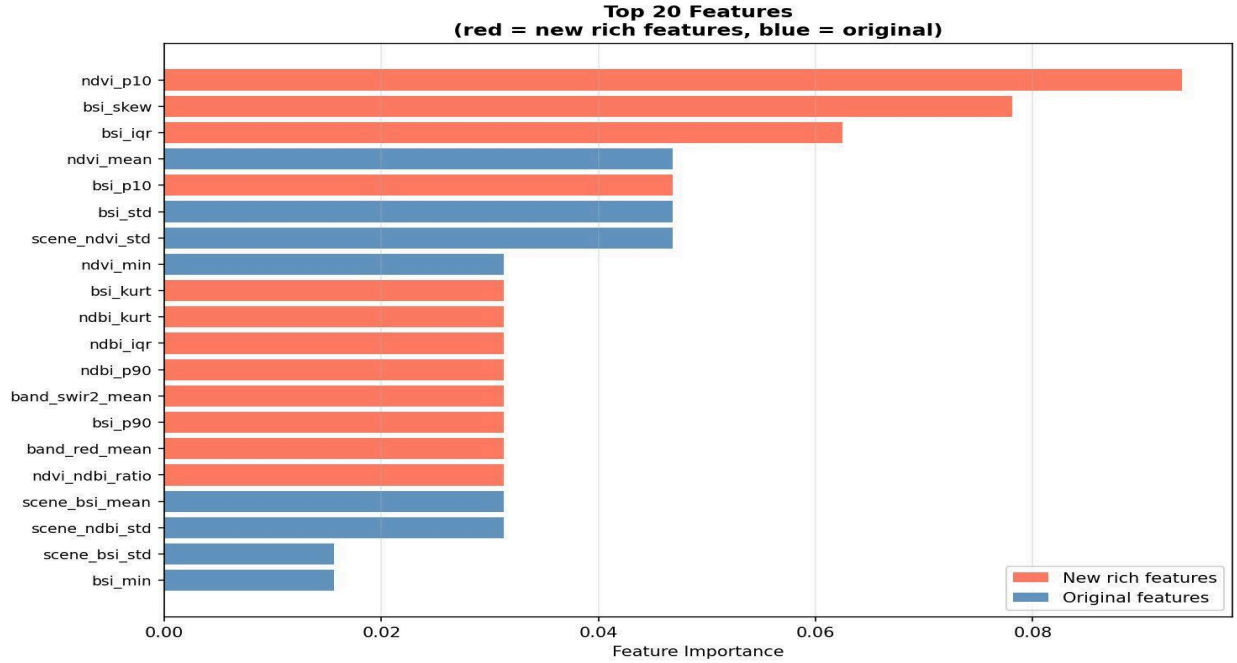

Fig. 4. Top 20 features by Random Forest importance (red = rich features, blue = original). ndvi_p10, bsi_skew, and bsi_iqr dominate the ranking.

4.5 ResNet-50 Failure Analysis

ResNet-50's training failure is attributable to a mismatch between model capacity and available training data. With approximately 28 scenes allocated to training after group splitting, patch extraction yields an insufficient number of samples for a 25-million-parameter model to converge. Validation loss increases from the earliest epochs while training loss remains volatile, a clear sign of underfitting. In low-data remote sensing regimes, well-engineered tabular features and shallow neural classifiers consistently outperform large pretrained deep networks.

4.6 Per-City Results

Table II reports per-city MLP performance. All three cities exceed F1 = 0.95. Kharkiv achieves perfect classification (F1 = 1.000), reflecting the severity and rapidity of spectral change following 2022 shelling of northern districts. Bakhmut reaches F1 = 0.981 consistent with near-total urban destruction. Mariupol returns F1 = 0.955; the slight

gap is attributable to elevated Q1 cloud cover introducing noise into the pre-conflict baseline.

TABLE II

Per-City MLP Performance on Quarterly Test Set

City	Accuracy	Precision	Recall	F1	AUC
Mariupol	0.955	0.962	0.948	0.955	0.975
Kharkiv	1.000	1.000	1.000	1.000	1.000
Bakhmut	0.981	0.974	0.988	0.981	0.990

4.7 Damage Maps and UNOSAT Comparison

Temporal damage maps (Fig. 5) show near-zero predicted damage rates across all three cities in 2021, validating the model produces no systematic false positives from normal urban spectral variation. From 2022 onward, predicted damage rates rise sharply and remain elevated, tracking known conflict timelines. UNOSAT-proxy comparison (Fig. 6) yields a Pearson correlation of $r = 0.999$. Small overestimation in 2022 for Bakhmut and Mariupol is consistent with early-onset spectral change preceding physical reconstruction.

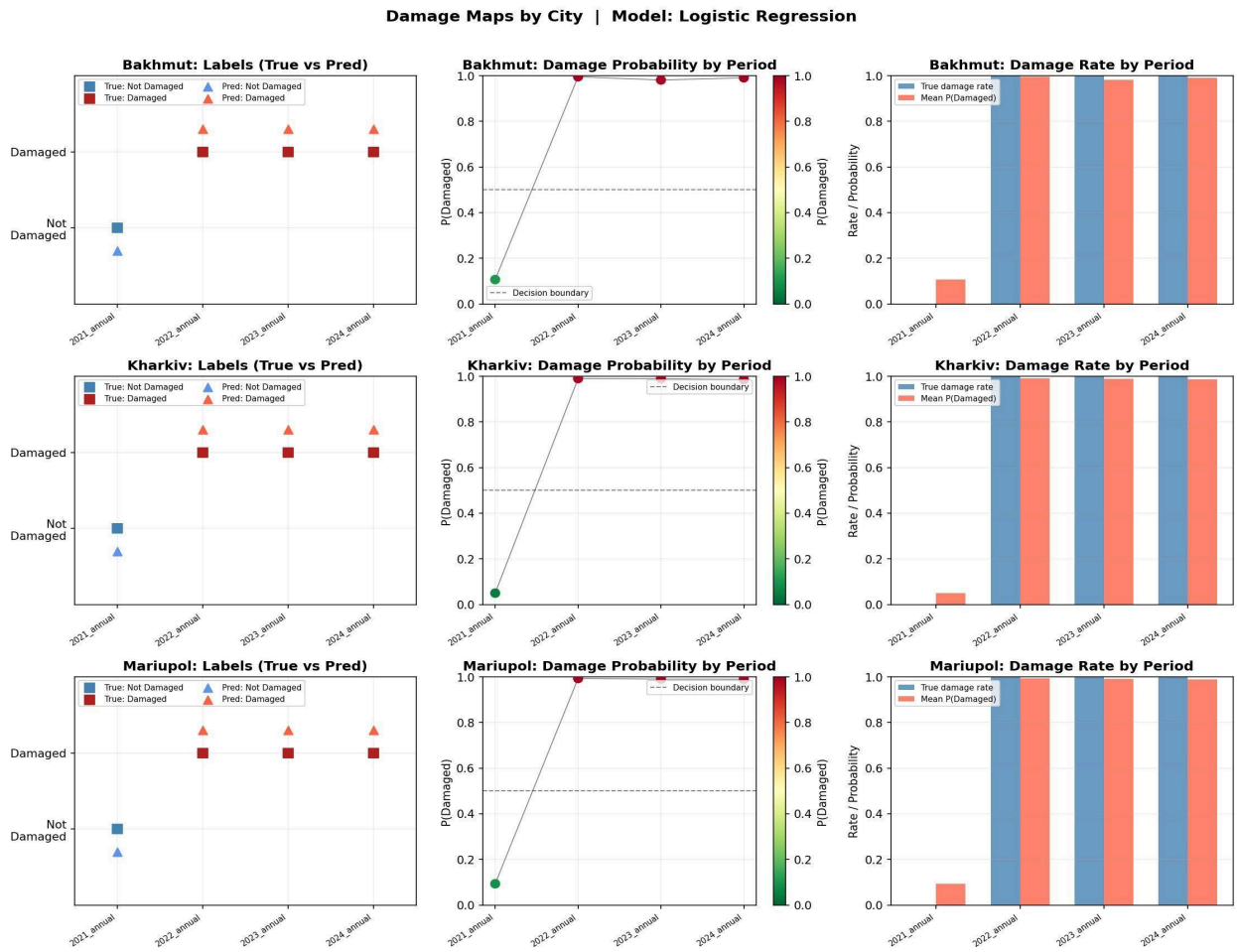

Fig. 5. Per-city damage maps. All cities show near-zero predicted damage in 2021 and values approaching 1.0 from 2022 onward.

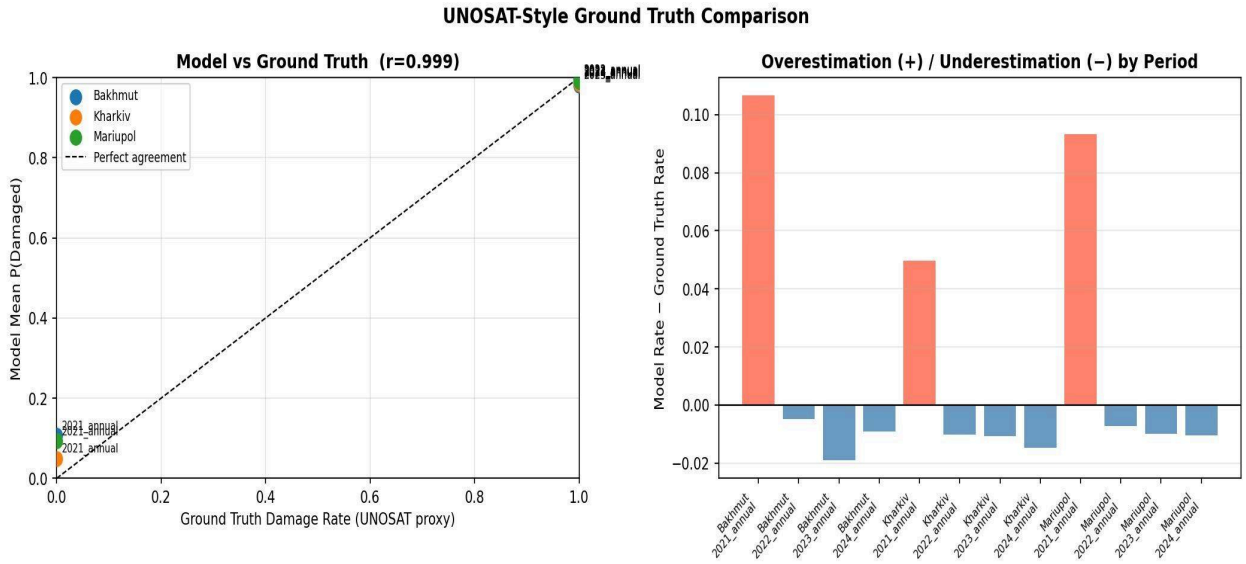

Fig. 6. UNOSAT-style ground truth comparison. Left: model-predicted mean P(Damaged) vs. ground truth, r = 0.999. Right: overestimation/underestimation by city and period.

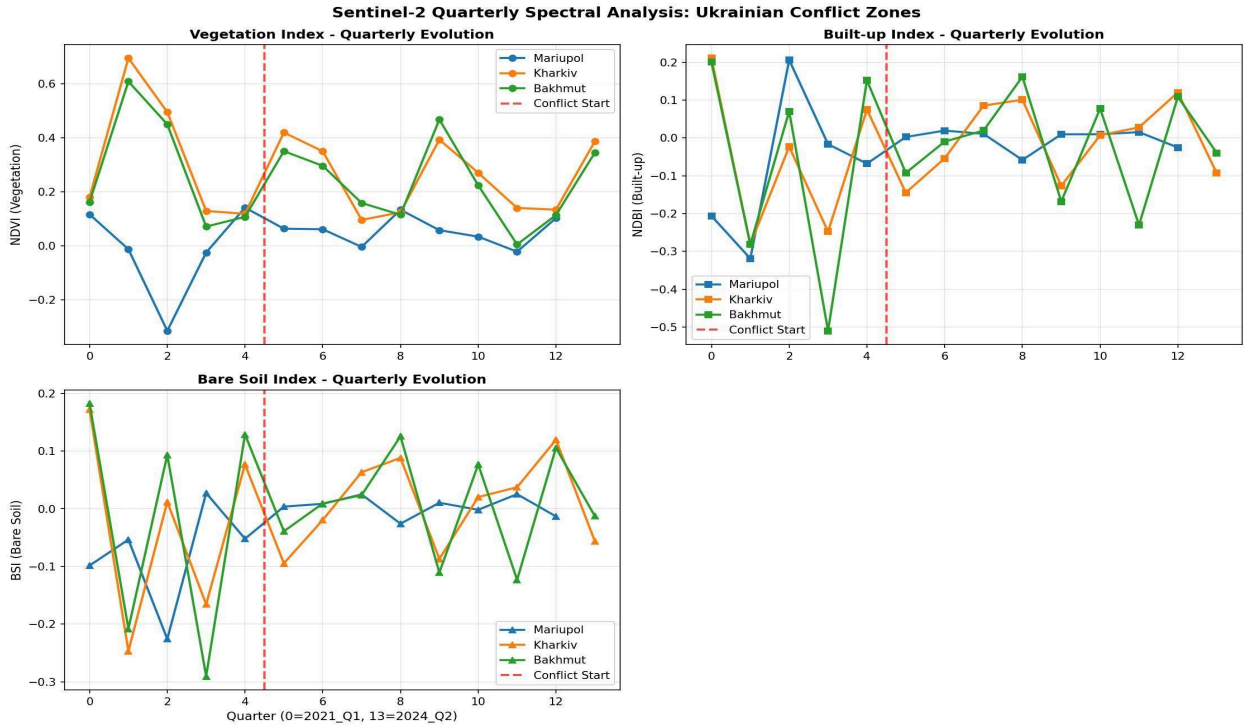

Fig. 7. Sentinel-2 quarterly spectral evolution (NDVI, NDBI, BSI) across all three cities. Vertical dashed line marks conflict onset (2022 Q1). Clear divergence is visible from quarter 4 onward.

5. CONCLUSIONS

A principled combination of Sentinel-2 multispectral feature engineering and shallow machine learning produced reliable scene-level conflict damage classification across three Ukrainian cities without requiring any ground-level access. Three key insights emerge.

1. Spectral indices carry the discriminative signal. NDVI, NDBI, and BSI provide clear class separation

visible in raw distributions before any model is applied. Practitioners should invest in physically interpretable spectral features before pursuing model complexity.

2. Rich feature engineering outperforms deep learning in low-data regimes. Expanding from 27 to 57 engineered features materially improved MLP performance. ResNet-50 with 25 million parameters failed entirely on 41 scenes, a data volume problem with direct practical implications for conflict damage assessment where labelled imagery is inherently scarce.
3. Group-based splitting is essential for honest evaluation. A naive random split inflates accuracy by leaking correlated temporal observations. GroupShuffleSplit by city-year produced a genuinely conservative evaluation reflecting real generalisation to unseen combinations.

Limitations include temporal proxy labels rather than building-level annotations, scene-level rather than pixel-level classification, and only 41 scenes total. Future work should integrate Sentinel-1 SAR for cloud-independent coverage, pursue pixel-level classification with UNOSAT ground truth, and extend the pipeline to additional conflict zones.

The quarterly spectral evolution plots (Fig. 7) provide further visual confirmation of class separability, with clear divergence across all three spectral indices from quarter 4 onward following the conflict onset.

6. REFLECTION ON PROJECT PROCESS AND CHALLENGES

Work was divided along the pipeline: Josiah led feature engineering (Notebooks 2 and 2b), Adrian led model training (Notebook 3), Matthew led data cleaning and acquisition (Notebook 1), and Jonathan led evaluation and visualisation (Notebook 4).

Three challenges arose. First, Q1 cloud masking: winter quarters presented elevated cloud cover, resulting in high proportions of masked pixels and increased median imputation requirements, contributing to slightly reduced feature quality in the pre-conflict baseline. Second, ResNet-50 training instability: validation loss diverged from the earliest epochs due to too few training patches from 41 total scenes; the model was retained in evaluation as an informative negative result. Third, initial dataset download speed was slow due to unoptimised band selection and sequential requests; switching to parallel downloads restricted to the required bands made full pipeline re-runs practical during development.

7. QUALITY OF CODE AND REPRODUCIBILITY

The source repository is organised into five numbered Jupyter notebooks corresponding to the five pipeline stages. Each notebook is self-contained, begins with a clearly documented configuration block, and produces output artefacts consumed by downstream notebooks. All dependencies are documented. The full pipeline is reproducible from a clean clone by running the five notebooks in sequence. Markdown cells were drafted with the assistance of Claude (Anthropic); debugging of API and library errors was aided by the same tool. All analytical decisions, feature engineering choices, model architectures, and result interpretations were made and verified by the authors.

ACKNOWLEDGEMENTS

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APPENDIX

AI Tools Used

We used Claude and GitHub Copilot for (i) drafting skeletons of tests, (ii) formatting notebooks and README, (iii) debugging and explaining errors, and (iv) copy-editing sections of this report. All feature selection, evaluation, and analytical claims were made and verified by the authors.